

Course Syllabus Part I

CSD 310 Database Development and Use

3 Credit Hours

Course Description

This course introduces the concepts of relational and non-relational database structures from a software developer perspective. Topics include defining key/value pairs, building table structures, manipulating records, using data types, and implementing access controls. The Structured Query Language (SQL) will be used to manipulate the models. Students will prepare a small application using a relational database management system.

Course Prerequisites

CSD 205 Introduction to Programming with Python

Course Skills

- Leverage creative thinking and innovative approaches to solve problems and overcome challenges in a development team.
 - Research stakeholder requirements, including documentation and coordination of specifications.
 - Apply information security techniques to protect the data being used in software applications.
 - Use the Python programming language to develop data-driven software applications.
 - Develop efficient SQL queries to improve the performance of software applications.
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Course Objectives

Students who successfully complete this course should be able to

1. Compare and contrast the differences between relational and non-relational database structures.
 2. Design data models to accurately reflect business requirements.
 3. Defend the use of non-relational database structures.
 4. Use the Python programming language to connect and interface with relational and non-relational database systems.
 5. Create, update, delete, and query records in relational and non-relational database systems.
 6. Apply industry best practices for managing access controls and data validation.
 7. Identify methods to mitigate security risks.
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Grading Scale

<u>Letter Grade</u>	<u>Percentage Grade</u>	<u>Letter Grade</u>	<u>Percentage Grade</u>
A	≥ 92.5%	C	< 76.5% and ≥ 72.5%
A-	< 92.5% and ≥ 89.5%	C-	< 72.5% and ≥ 69.5%
B+	< 89.5% and ≥ 86.5%	D+	< 69.5% and ≥ 66.5%
B	< 86.5% and ≥ 82.5%	D	< 66.5% and ≥ 62.5%
B-	< 82.5% and ≥ 79.5%	D-	< 62.5% and ≥ 59.5%
C+	< 79.5% and ≥ 76.5%	F	< 59.5%

Topic Outline

- I. Core Principles
 - a. Business Rules
 - b. Data Modeling
 - c. Relational vs. Non-Relational Databases
 - d. Normalization
 - e. Denormalization
- II. Non-Relational Databases
 - a. JSON
 - b. Collections
 - c. Creating, Updating, and Deleting Documents
 - d. Collection Queries
 - e. Indexes, Validation, and Access Control
- III. Relational Databases
 - a. SQL
 - b. Tables
 - c. Creating, Updating, and Deleting Records
 - d. Table Queries
 - e. Advanced Queries
 - f. Validation and Access Controls

This syllabi update reflects grading scale policy updates effective 4/1/2024.